Polynomials — problem 10.82

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It is known of a polynomial over $\mathbb{Z}$ that $p(n) > n$ for every positive integer n. Consider $x_1 = 1$, $x_2 = p(x_1) \ldots$. We know that, for any positive integer N, there exists a term of the sequence divisible by N. Prove that $p(x) = x + 1$.